

JAYANTH DASAMANTHARAO

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PROFESSIONAL EXPERIENCE

Applied AI Engineer, EliteUS Software Solutions Aug 2024 - Present | Edison, NJ

- Architected and deployed agentic LLM systems (RAG + tool-using agents) powering real customer workflows, serving 12,500+ users with production-grade reliability.
- Designed multi-step reasoning agents with function calling, retrieval augmentation, reranking, and memory mechanisms to automate decision workflows end-to-end.
- Owned full ML lifecycle: model selection, prompt/adaptor tuning, evaluation frameworks (precision/recall, BLEU, human review), A/B testing, monitoring, and iterative optimization in AWS/Azure.
- Built scalable ML microservices and REST APIs (FastAPI), implementing authentication, guardrails, observability, telemetry, and failure recovery strategies for production AI systems.
- Designed automated evaluation pipelines and feedback loops to improve agent accuracy in production.

Data Science Intern, Harvest Software Solutions June 2023 - Sept 2023 | Jacksonville, Florida

- Engineered scalable ML pipelines and monitoring systems for large advertising datasets, increasing campaign visibility from 15% to 45% while ensuring reliability and performance.
- Built ML services and NLP APIs using XGBoost, H2O AutoML, and FastAPI, improving sentiment classification accuracy by 65% and campaign spend efficiency by 50%.
- Designed SQL/Python data processing and evaluation pipelines for feature engineering, model training, and validation, enabling rapid experimentation and improving engagement quality by 40%.

Application Development Associate, Accenture June 2021 - Aug 2022 | Hyderabad, India

- Developed and optimized Python-based ETL and data processing pipelines using hash maps, sets, indexed lookups to efficiently clean, deduplicate, and transform large datasets, reducing production issues by 50%.
- Implemented SQL workflows and joins with partitioning, indexing, and window functions, enabling efficient aggregation and analysis over millions of records in AWS Athena and Redshift.
- Applied batching, caching, and algorithmic optimizations for data validation and pipeline execution, improving system performance and reliability by 25%.

TECHNICAL SKILLS

Programming:	Python, R, SQL. Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, FastAPI.
Machine Learning :	Model Fine-tuning, evaluation and experimentation, deployment, Feature Engineering.
GenAI & LLMs:	LLM-powered applications, RAG pipelines, vector databases, prompt engineering, LLM evaluation.
AI Platforms/Agents:	Agentic systems, GenAI systems, memory architectures, tool calling, guardrails.
APIs & Services:	REST APIs, API versioning, authentication, developer tooling, ETL pipelines, distributed processing.
Cloud & MLOps:	PostgreSQL, Data Warehousing, AWS (Bedrock, Athena, Redshift), GCP, Azure, Vectordb.

EDUCATION

Rutgers University, Master of Science in Data Science Sept 2022 - May 2024 | New Brunswick, NJ

Personal Website: <https://jayanthdasamantharao.github.io/>

Andhra University B.Tech in Electrical and Electronics Engineering 2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Production Agentic RAG Platform: Built and deployed a production agentic retrieval and generation system integrating embeddings, tool use, memory, evaluation loops, and monitoring to improve response accuracy, reliability, and latency.

Self-Driving Cars (Computer Vision): Built autonomous driving cars with vision pipeline using CNNs, YOLO, Faster R-CNN, achieving 95%+ object detection accuracy.

Bayesian Classification: Implemented Bayesian logistic regression in PyMC3 to predict diabetes status, incorporating informative priors, achieving 0.85 F1-score.

For additional projects and code samples: <https://github.com/JayanthDasamantharao>